Poisson problem with Dirac delta source

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Consider the following problem on $\Omega = (-1, 1)$:

$$\begin{cases} -u'' = \delta_0, \\ u'(-1) = u'(1) = 0. \end{cases}$$

Using physical intuition, we expect u to be a "tent" function with a peak at $x = 0$ and linear decay towards the boundaries. However, we want to rigorously analyze this problem using functional analysis.

So, we are looking for a solution u which is continuous and with derivatives in L^2. This suggests that we should work in the Sobolev space $H^1((-1, 1))$.

Following the weak formulation of the problem, we have by the corollary of Lax-Milgram that

$$\|u\|_{H^1} \leq C\|\delta_0\|_{(H^1)^*}.$$

So, we expect δ_0 to be a bounded linear functional on $H^1((-1, 1))$.

Check boundness of δ_0 on $H^1((-1,1))$ 1/n

Let's compute the norm of δ_0 in the dual space $(H^1)^*$. First, recall that $H^1((-1,1))$ is continuously embedded in $C^0([-1,1])$, which means that every function in H^1 is continuous and can be evaluated at points.

$$\|\delta_0\|_{(H^1)^*} = \sup_{u \in H^1, u \neq 0} \frac{|\delta_0(u)|}{\|u\|_{H^1}} = \sup_{u \in H^1, u \neq 0} \frac{|u(0)|}{\|u\|_{H^1}}.$$

Now, let's study $|u(0)|$ and let's try to bound it by the $\|u\|_{H^1}$.
By the fundamental theorem of calculus, we have for any $y \in (-1,1)$:

$$u(0) = u(y) + \int_y^0 u'(x)\,dx.$$

We can average over y in $(-1,1)$ to get the following line, then using Cauchy-Schwartz we get the result

$$u(0) = \frac{1}{2}\int_{-1}^1 u(y)\,dy + \frac{1}{2}\int_{-1}^1 \int_y^0 u'(x)\,dx\,dy = \frac{1}{2}\int_{-1}^1 u(y)\,dy + \int_y^0 u'(x)\,dx.$$

$$|u(0)| \leq \frac{1}{2}\sqrt{2}\|u\|_{L^2} + \sqrt{2}\|u'\|_{L^2} \leq C\|u\|_{H^1}.$$

Boundness of δ_0 on $H^1((-1,1))$ and Riesz representation

So, δ_0 is indeed a bounded linear functional on $H^1((-1,1))$. This means that we can apply the Riesz representation theorem to find a function $v_{\delta_0} \in H^1((-1,1)) \subset C^0([-1,1])$ such that for all $u \in H^1((-1,1))$:

$$\delta_0(u) = u(0) = \langle u, v_{\delta_0} \rangle_{H^1} = \int_{-1}^{1} u(x) v_{\delta_0}(x)\, dx + \int_{-1}^{1} u'(x) v'_{\delta_0}(x)\, dx.$$

Goal: find $v_{\delta_0} \in C^1([-1,1])$ that verify the above equation.

Weak Formulation

Integrate by parts:

$$\int u' v'_{\delta_0} = -\int u v''_{\delta_0} + u(1) v'_{\delta_0}(1) - u(-1) v'_{\delta_0}(-1).$$

Hence, we have that

$$\int_{-1}^{1} u(x)(v_{\delta_0}(x) - v''_{\delta_0}(x))dx + u(1) v'_{\delta_0}(1) - u(-1) v'_{\delta_0}(-1) = u(0).$$

Then, for any $u \in C_c^1(\Omega)$:

$$u(0) = \int_{-1}^{1} u(v_{\delta_0} - v''_{\delta_0})dx.$$

This means that we can construct a sequence of $u_\varepsilon \in C_c^1(\Omega)$ such that $u_\varepsilon \to \tilde{u}$ where $\tilde{u}(1) = 1$ and $\tilde{u}(-1) = 0$. Then, we have that

$$\lim_{\varepsilon \to 0} \int_{-1}^{1} u_\varepsilon (v_{\delta_0} - v''_{\delta_0})dx - \tilde{u}(0) = \lim_{\varepsilon \to 0} \int_{-1}^{1} u_\varepsilon (v_{\delta_0} - v''_{\delta_0})dx - \tilde{u}(0) + v'_{\delta_0}(1).$$

This means that $v'_{\delta_0}(1) = 0$. Similarly, we can construct a sequence of u_ε such that $\tilde{u}(1) = 0$ and $\tilde{u}(-1) = 1$ to get $v'_{\delta_0}(-1) = 0$.

So in distributional sense we want to solve:

$$v_{\delta_0} - v''_{\delta_0} = \delta_0.$$

Boundary Conditions

So we solve the boundary value problem:

$$\begin{cases} v_{\delta_0} - v''_{\delta_0} = \delta_0, \\ v'_{\delta_0}(-1) = v'_{\delta_0}(1) = 0. \end{cases}$$

Solve the ODE

For $x \neq 0$:

$$v'' - v = 0$$

General solution:

- $x > 0$: $Ae^x + Be^{-x}$
- $x < 0$: $Ce^x + De^{-x}$

Conditions:

- continuity at 0
- jump:

$$\lim_{\varepsilon \to 0} \int_{-\varepsilon}^{\varepsilon} v'' - v \,\mathrm{d}x = -\int_{-\varepsilon}^{\varepsilon} \delta_0 = -1,$$
$$\lim_{\varepsilon \to 0} v'(\varepsilon) - v(-\varepsilon) - 2\varepsilon v(0) = v'(0^+) - v'(0^-) = -1.$$

Solve the linear system

General solution:

- $x > 0$: $Ae^x + Be^{-x}$
- $x < 0$: $Ce^x + De^{-x}$

$$\begin{cases} Ae^1 - Be^{-1} = 0 & \text{(Neumann BC at } x = 1\text{)} \\ Ce^{-1} - De^1 = 0 & \text{(Neumann BC at } x = -1\text{)} \\ A + B = C + D & \text{(continuity at } x = 0\text{)} \\ A - B - (C - D) = -1 & \text{(jump condition at } x = 0\text{)} \end{cases}$$

Solution is

$$\begin{cases} A = \frac{e^{-1}}{e^1 - e^{-1}} \\ B = \frac{e}{e^1 - e^{-1}} \\ C = \frac{e}{e^1 - e^{-1}} \\ D = \frac{e^{-1}}{e^1 - e^{-1}} \end{cases}$$

Final Result

$$v_{\delta_0}(x) = \frac{\cosh(1 - |x|)}{2\sinh(1)}$$

- ✓ $v_{\delta_0} \in H^1((-1, 1))$
- ✓ Represents the Dirac delta
- ✓ One can check that all the above conditions are statisfied

Key Insight

In 1D:

$$H^1 \subset C^0$$

So:

- δ_0 is bounded in $(H^1)^*$
- Riesz applies

Higher Dimensions

In $ \mathbb{R}^2, \mathbb{R}^3, \dots$:

$$H^1 \not\subset C^0$$

So:

- δ_0 is **not bounded** in $(H^1)^*$
- No Riesz representative in H^1